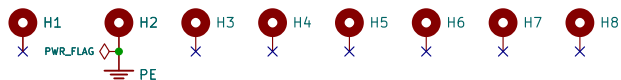

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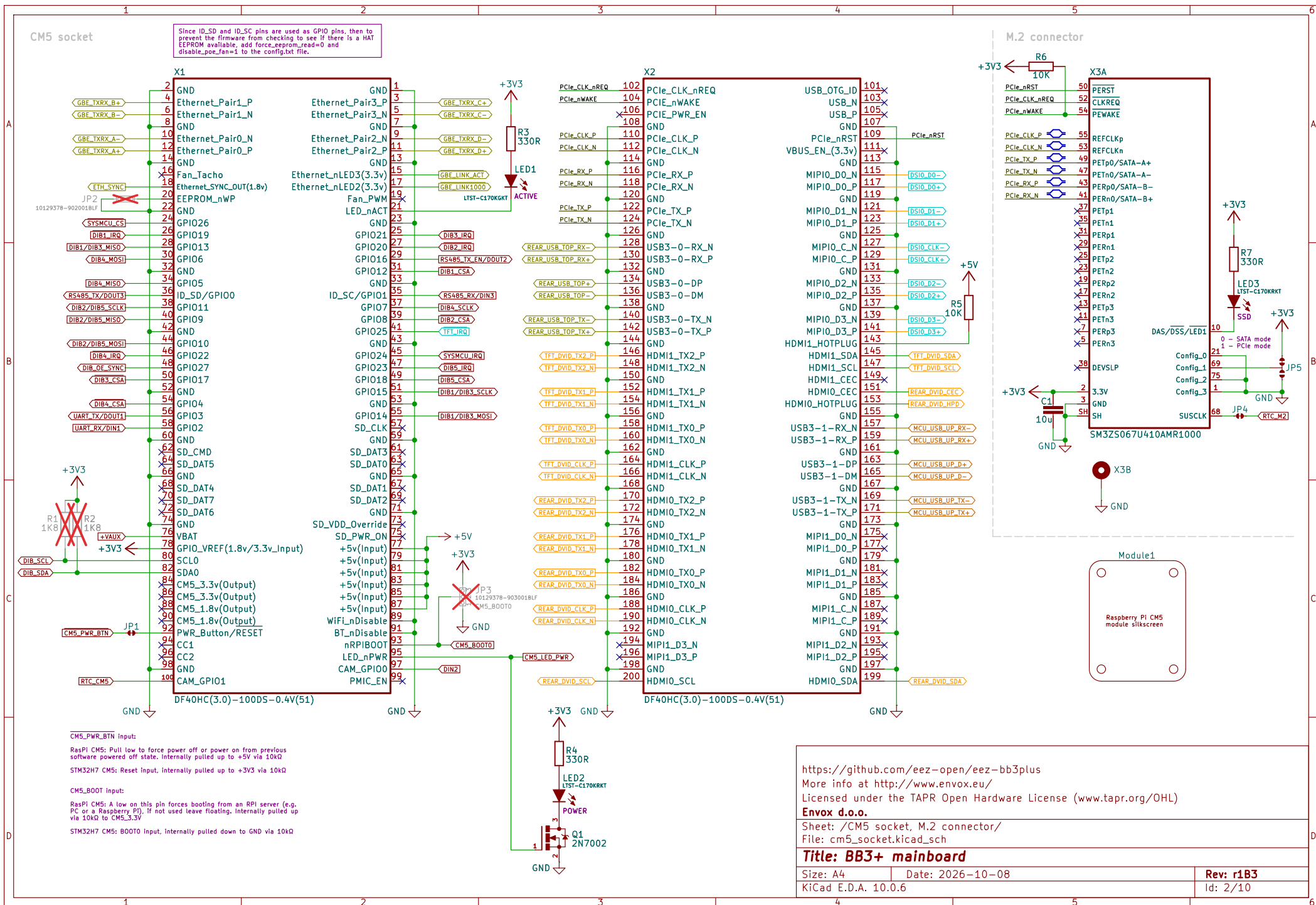
USB Port	USB type	STM32H7 CM5	Raspberry PI CM5
Rear top	USB A	FS only	USB #0, HS/SS
Rear bottom	USB A	FS/HS	USB Hub PI Flex
Front	USB C	USB Hub P2, HS/SS, Dual role	
DIB #1	DIB 2.0	USB Hub P7, HS/SS	
DIB #2	DIB 2.0	USB Hub P6, HS/SS	
DIB #3	DIB 2.0	USB Hub P5, HS/SS	
DIB #4	DIB 2.0	USB Hub P4, HS/SS	
DIB #5	DIB 2.0	USB Hub P3, HS/SS	

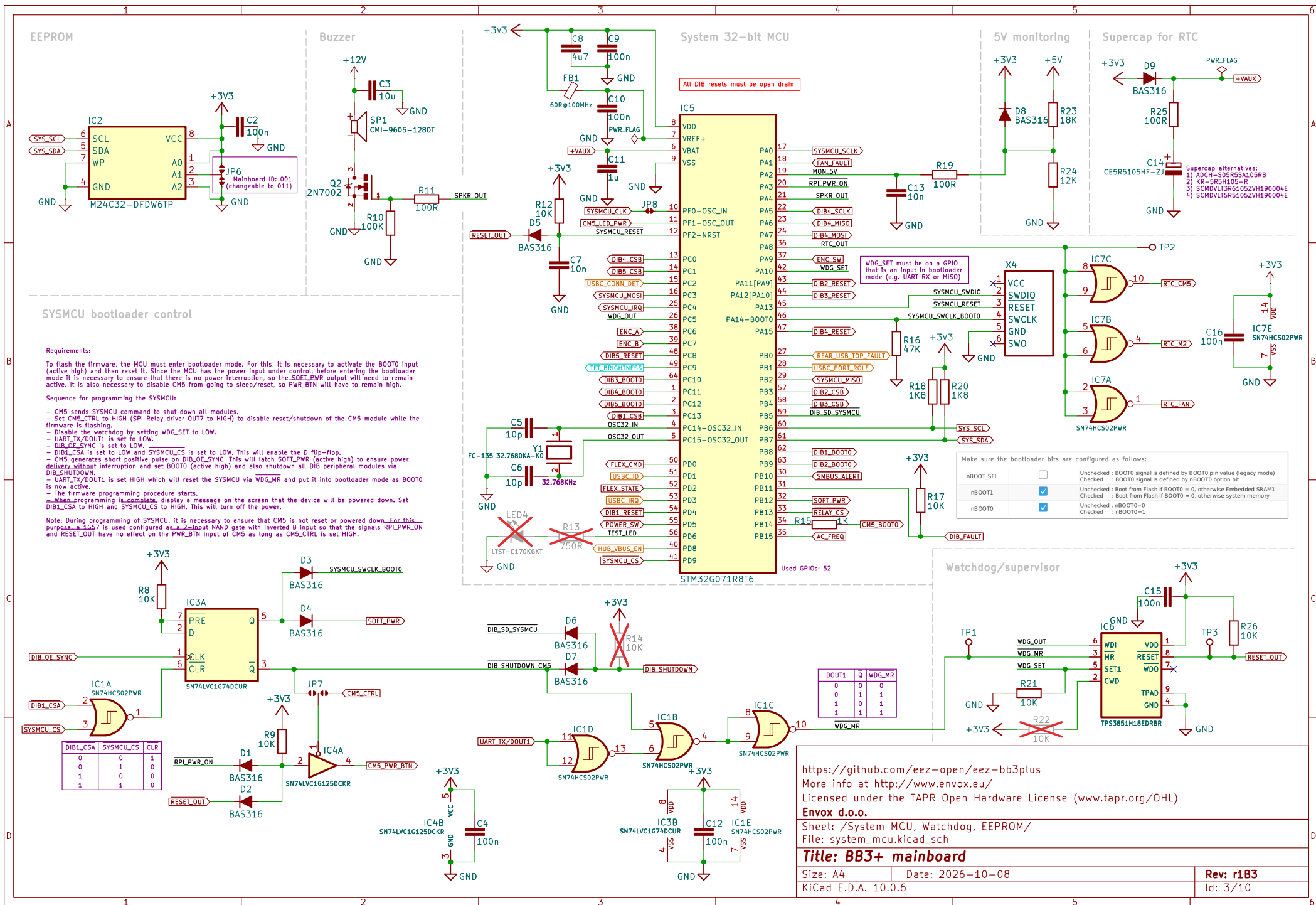
I2C address	Device
0x51	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x7A	PTN5150A USB CC controller

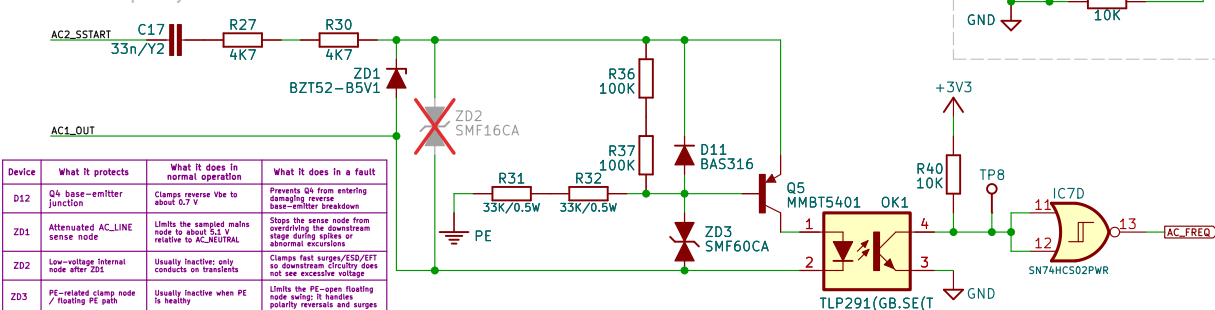
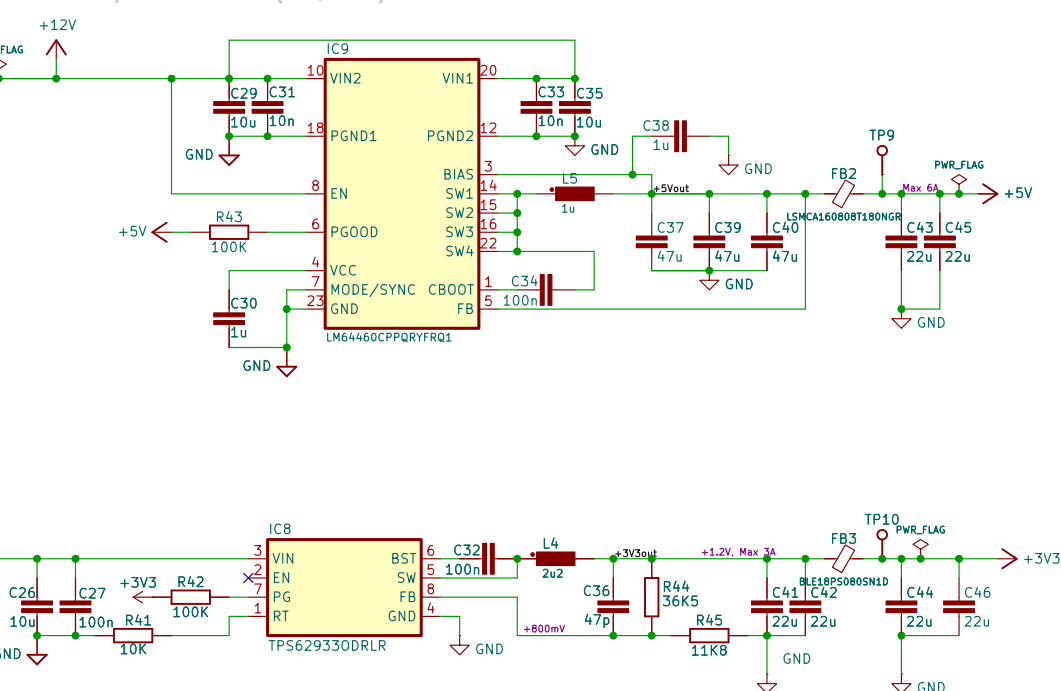
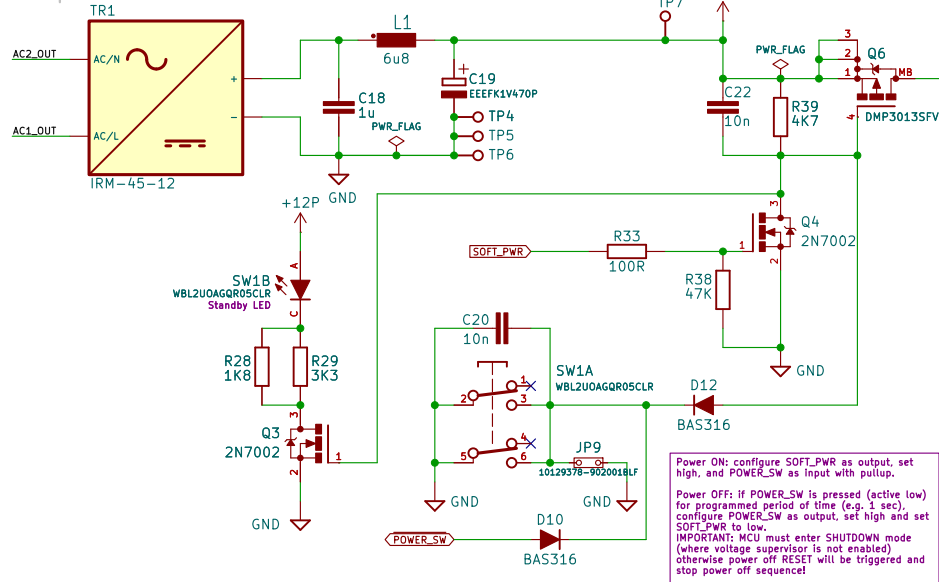
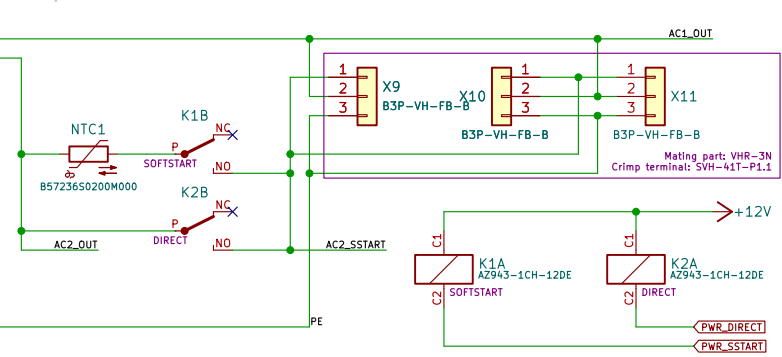
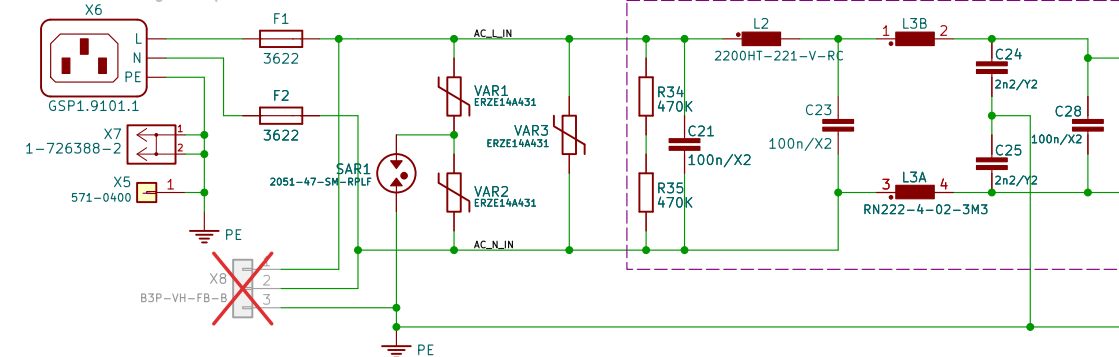


Title: BB3+ mainboard

Id: 1/10



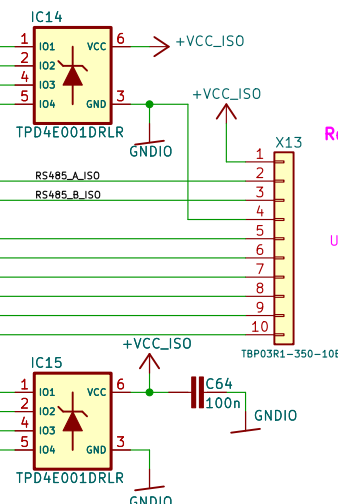
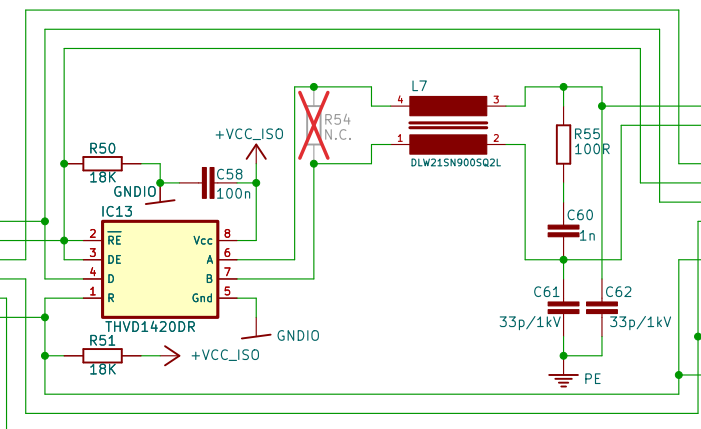
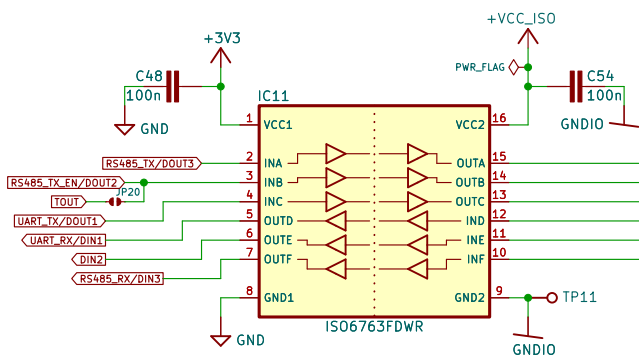




Device	What It protects	What it does in normal operation	What it does in a fault
D12	Q4 base-emitter junction	Clamps reverse Vbe to about 0.7 V	Prevents Q4 from entering damaging reverse base-emitter breakdown
Z01	Attenuated AC LINE sense node	Limits the sampled mains node to about 5.1 V relative to AC_NEUTRAL	Stops the sense node from overdriving the downstream stage during spikes or abnormal circumstances
Z02	Low-voltage Internal node after Z01	Usually inactive; only conducts on transients	Clamps fast surges/ESD/ESR to downstream circuitry does not see excessive voltage
Z03	PE-related clamping node / floating PE path	Usually inactive when PE is healthy	Limits the PE-opp floating node during fault handling (polarity reversals and surges

Id: 4/10

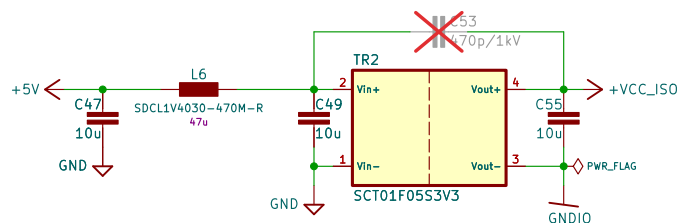
Isolated digital I/O & RS485 transceiver (half-duplex)



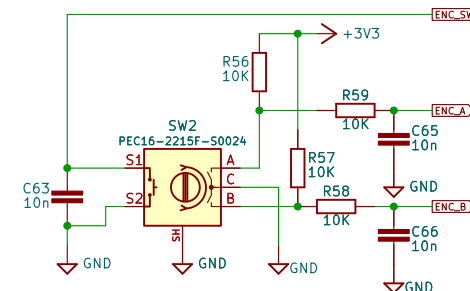
Rear panel I/O connector

- +VCC_ISO 1
- RS485_A_ISO 2
- RS485_B_ISO 3
- GND_ISO 4
- UART_TX/DOUT1 5
- DOUT2 6
- DOUT3 7
- UART_RX/DIN1 8
- DIN2 9
- DIN3 10

Isolated regulated power supply (+5V)

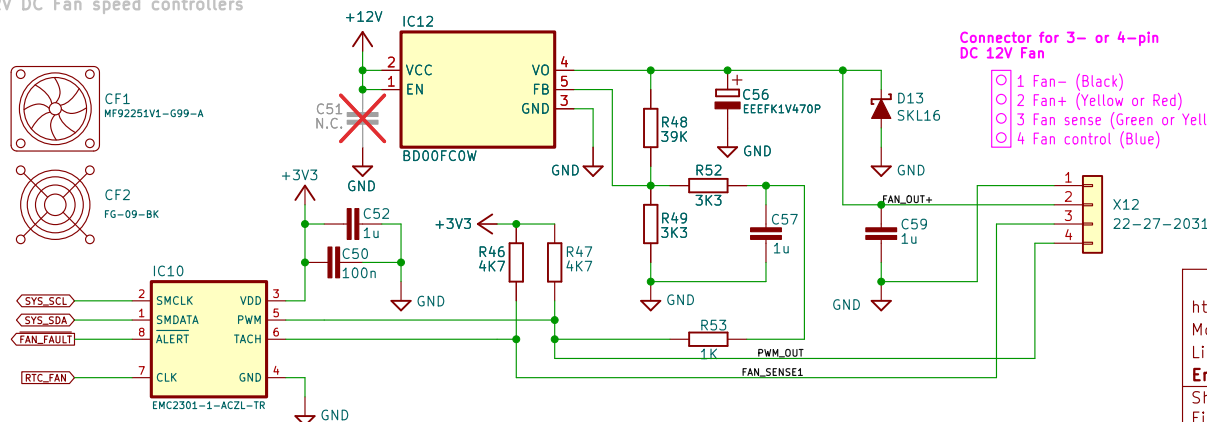


Encoder with switch



Encoder shaft length: 15 mm / Flatted
 Knob head Ø31mm A1031648 (OKW) + Knob pointer A1101007 (OKW)
 Optional: Knob head Ø31mm A3131068 (OKW) + Knob cap A3231007 (OKW)
 Knob head Ø24mm A1024648 (OKW) + Knob pointer A1101007 (OKW)

12V DC Fan speed controllers



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

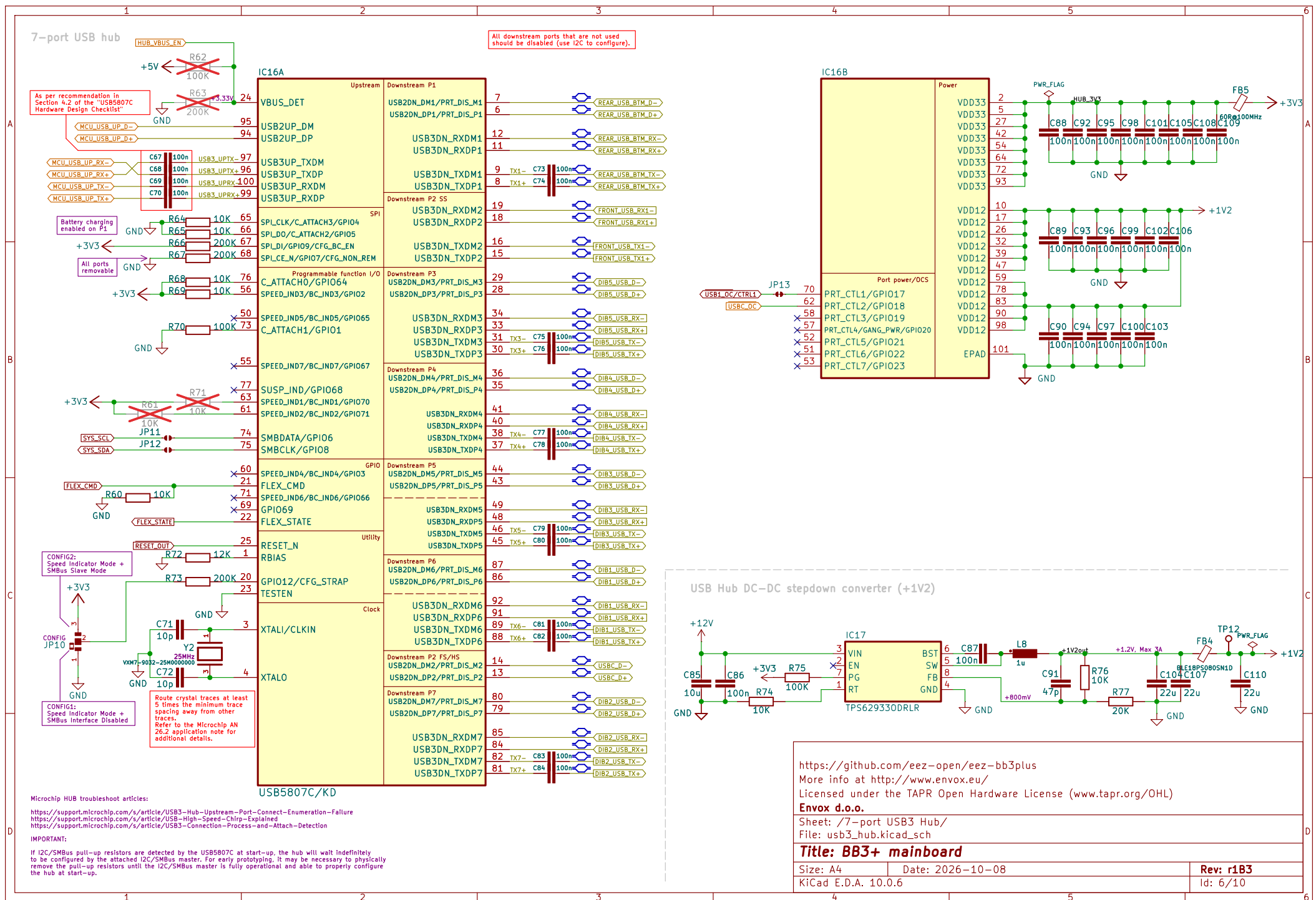
Title: BB3+ mainboard

Size: A4 Date: 2026-10-08

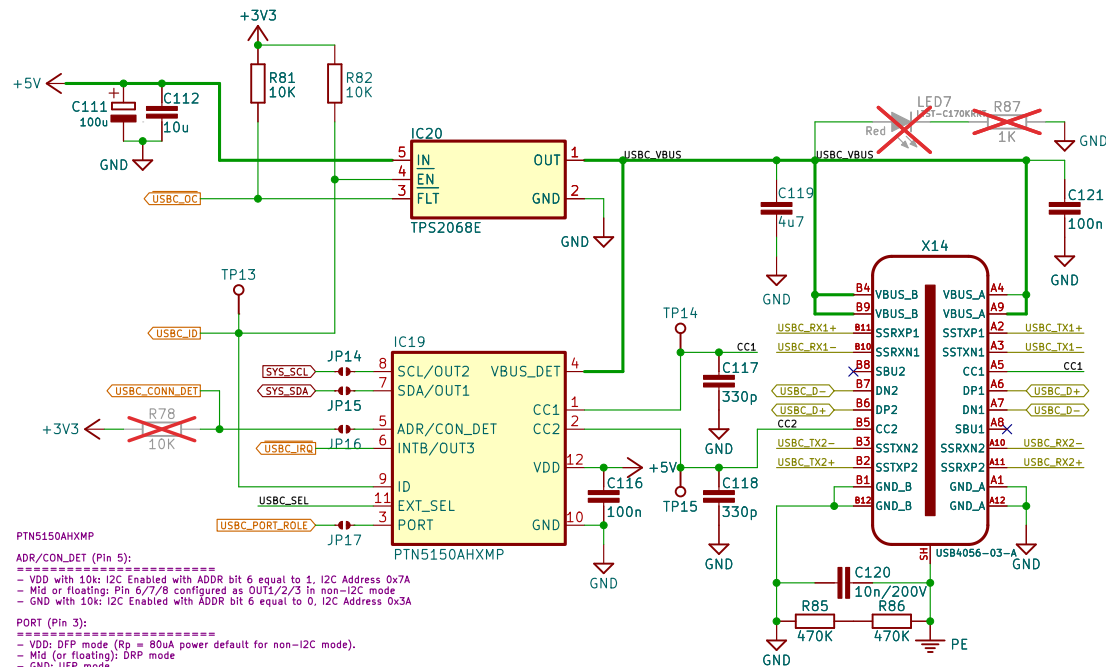
KiCad E.D.A. 10.0.6

Rev: r1B3

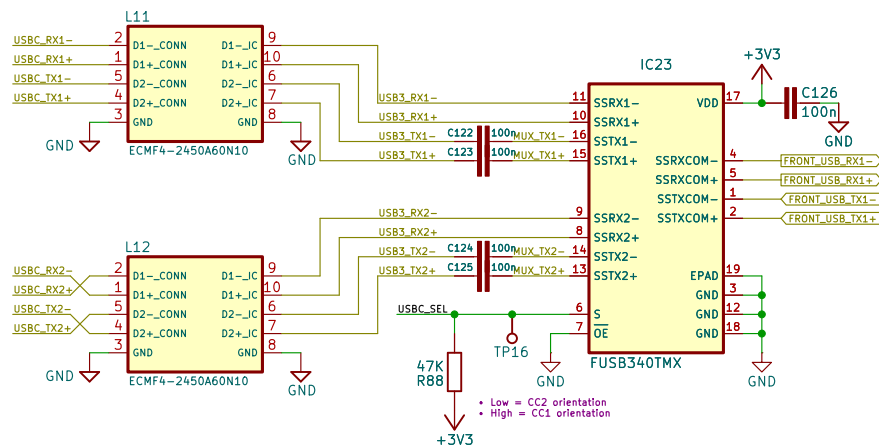
Id: 5/10



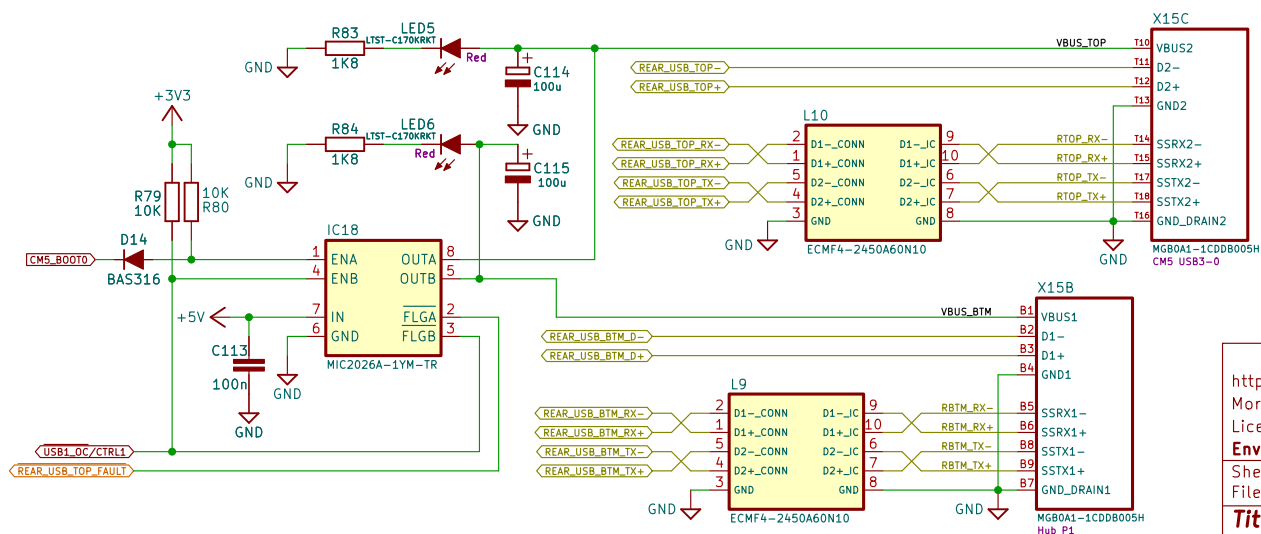
USB-C 3.0 front port (source max 1.5A)



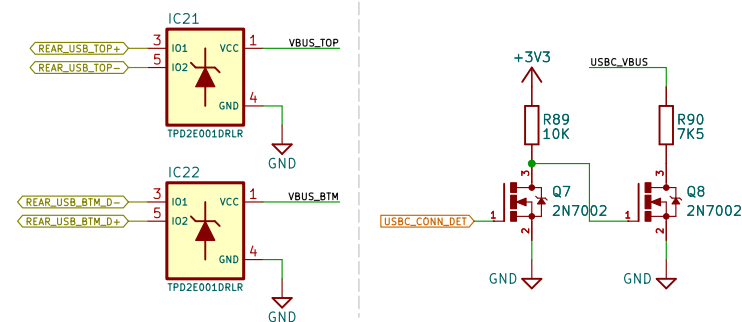
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



VBus Discharge Circuit



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Rear USB, USB-C/

File: usb_conn.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.6

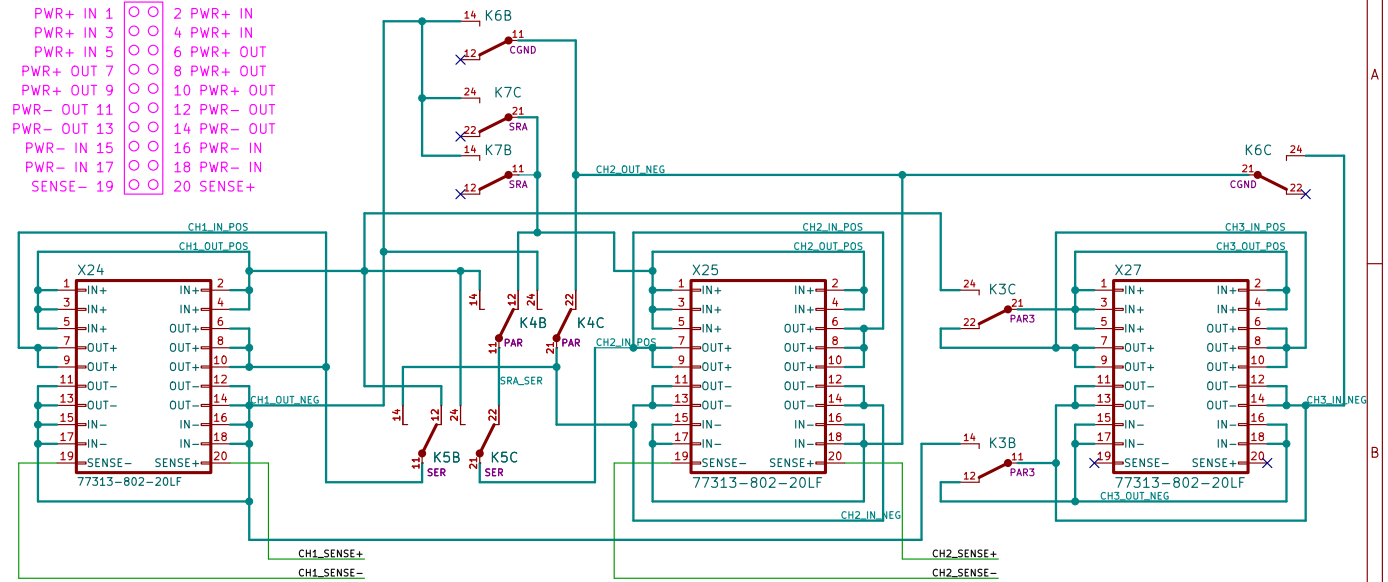
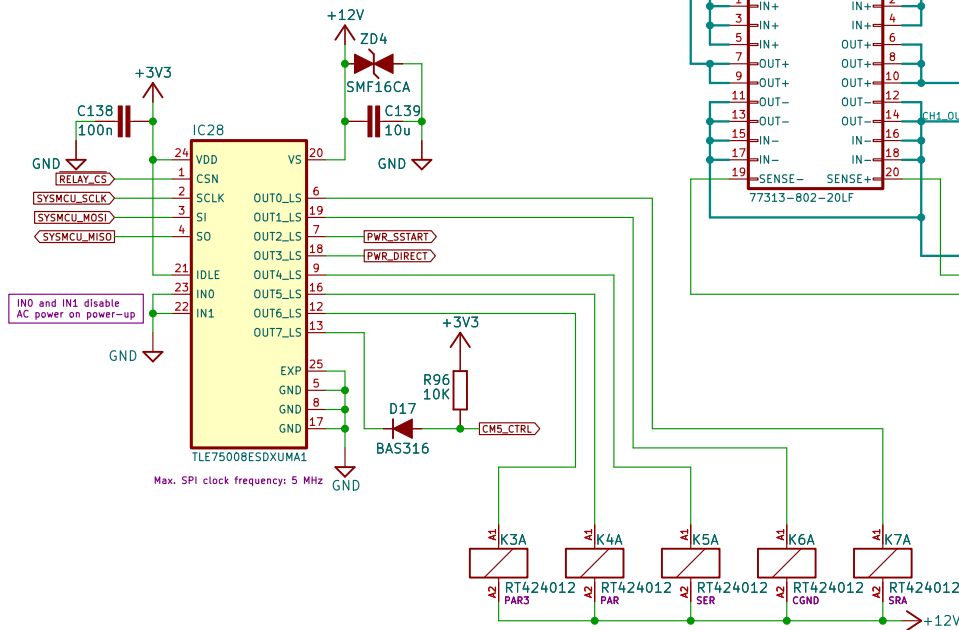
Rev: r1B3

Id: 7/10

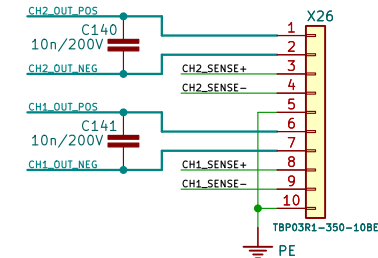
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ OUT
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	2
CH2_OUT- 3	4
CH2_SENSE+ 5	6
CH2_SENSE- 7	8
PE 9	10
CH1_OUT+ 11	12
CH1_OUT- 13	14
CH1_SENSE+ 15	16
CH1_SENSE- 17	18
PE 19	20

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: //Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 10/10